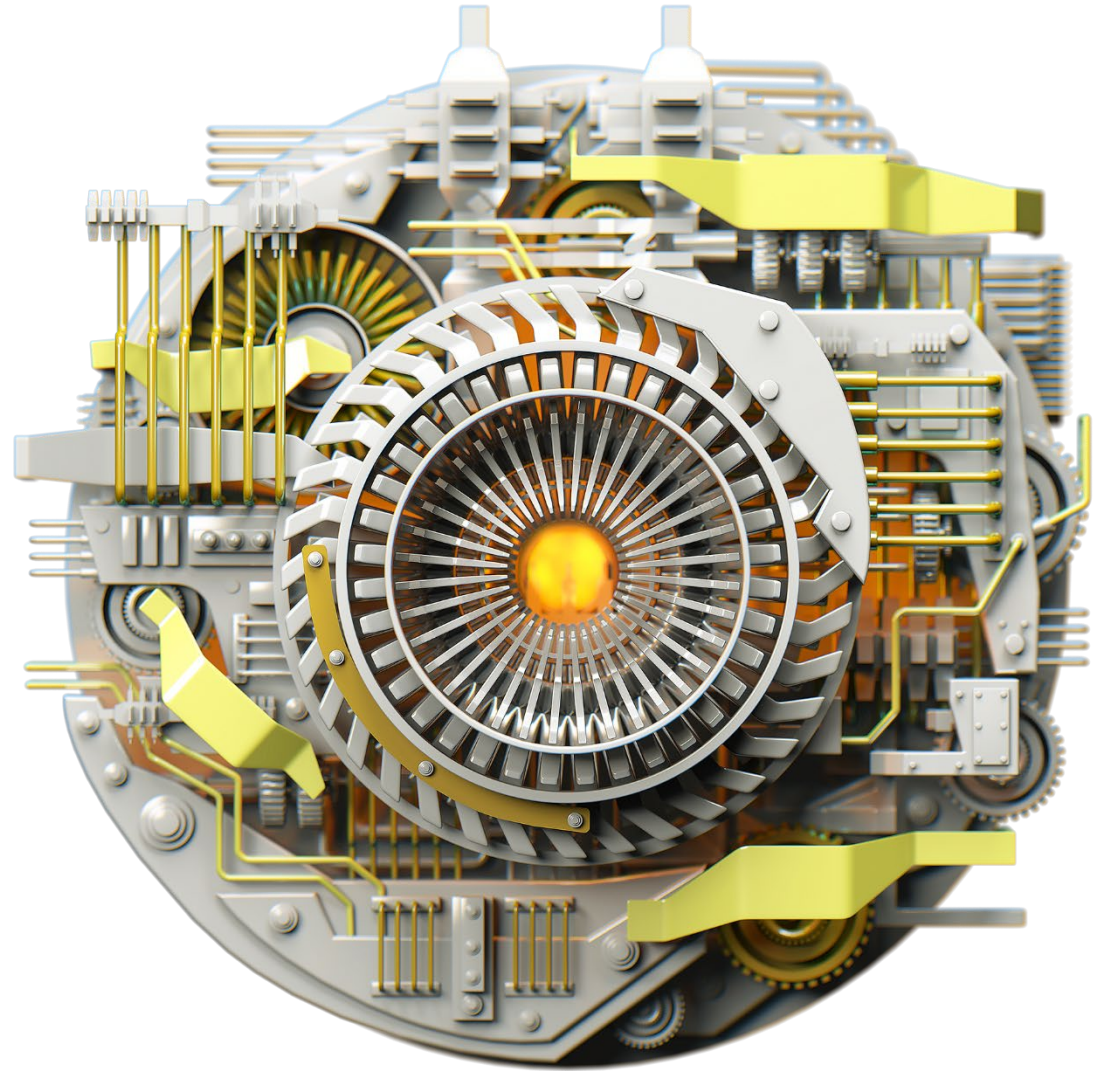




OCTOBER 11TH, 2024

Gen AI - API Testing Tool



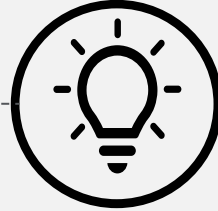
API Testing Tool – Accelerating API Quality



Problem Statement

Limitations with traditional way of API testing:

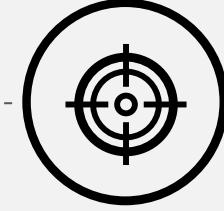
- **Manual effort** – Creating test data manually is time-consuming & prone to human error.
- **Limited coverage** – Test coverage is limited to scenarios explicitly thought of by developers, potentially missing critical paths.
- **Manual updates required** – When an API changes, test data must be manually updated to reflect the new structure & functionality.
- **Scalability issues** – As the project grows, managing and maintaining large number of APIs & test data becomes increasingly difficult and resource-intensive.



Solution

An API testing tool build on Python which generates test datasets using Gen AI with the help of two endpoints:

- **/upload** – This endpoint accepts swagger file and generate negative, positive & runtime test datasets based on the swagger documentation using **Open AI**.
- **/execute** – The datasets received as response from the upload endpoint, when fed to this endpoint, executes your APIs with the datasets and returns the response.



Impact

Impact of API testing tool:

- **Improved coverage with minimal efforts** – Broader test coverage with less effort, 40 % reduction in effort to achieve better coverage
- **Consistency & Reduced maintenance** – Ensures that test data remain aligned with current state of API and significantly lowers ongoing maintenance burden.
- **Scalable test management** – Automates the generation of test data & execution of APIs, easily scaling with complexity & size of project.
- **Alignment with requirements** – Tests are more closely aligned with business requirements, ensuring software meets its intended goals.



Global Property & Casualty Insurance Provider

Gen AI based API Testing

SITUATION

A global Insurance provider undergoing digital transformation journey, has **30+ Java microservices** providing integration with the backend data sources and other core systems. With the **evolving requirements and scale of the API's**, there is a **constant challenge to maintain the quality with minimum manual efforts**.

SOLUTION

- **API testing tool** that generates positive, negative, and runtime test datasets based on **Swagger documentation using Open AI**, and can execute API endpoints with these datasets to provide responses
- It enhances test efficiency with reusable regression suites and preserves various test datasets
- It also offers a comprehensive test strategy covering end-to-end capabilities, including corner cases, regression testing, and test data management

IMPACT

ESTIMATED NET REDUCTION
OF ...

35%-40%

IN AUTOMATION TESTING EFFORTS – 4 HOURS SAVED FOR TEST SCRIPT, TEST DATA GENERATION, EXECUTION AND RUN REPORT FOR A MICROSERVICE CONTAINING 30 APIs, WHICH TRADITIONALLY TOOK 10 HOURS

IMPROVED TEST COVERAGE OF ...

50%

IN REDUCED EFFORTS COMPARED TO SCRIPTED SCENARIOS DUE TO BROADER RANGE OF AUTO GENERATED TESTS, INCLUDING CORNER CASES AND UNEXPECTED INPUTS

Postman API testing vs Gen AI API Testing Tool

A comprehensive comparison between Postman based API Testing and Gen AI based API Testing on the various important parameters like efforts, coverage, efficiency



Postman based API Testing



Gen AI – API Testing tool

Testing approach	- Implement test cases for API execution using scripts, for generating test data for positive and negative scenarios - Execute test cases using scheduled iterations	GUI based tool generates positive, negative, and runtime test datasets based on Swagger documentation using Open AI, and can execute API endpoints with these datasets to provide responses
Testing efforts	An average of 10~ hours for limited test data creation, iteratively execute the test cases and generate test execution report	An average of 6~ hours to document strong Swagger specifications for the API endpoints, generate test data and execute the test cases
Test Coverage	Limited based on the manual combinations of positive and negative data generation scripts implemented	Extensive positive, negative and run time test data sets are generated based on the configuration provided
Test execution report	Need to manually collate all the positive and negative test results to generate a test execution report	A Python agent will calculate the positive and negative test results and provide the report for download
Testing Efficiency		
Ease of use	Requires some technical expertise to implement the multi step process of generating the test data and execute the test cases using the tool	A GUI provides simple options to upload the documentation and run the complete process at a click of a button
Technical Highlights	Postman API execution tool can be used to automate the testing of endpoints - This includes implementing scripts to generate test data and scheduling the iterations	- Built using Python which uses Open API to generate test data - Agent built using Python to execute test cases and generate test report

Steps to setup Gen AI API Testing Tool

Below are the prerequisites and steps to download the latest code from Deloitte Enterprise Github and setup the tool on your local

Pre-requisites :

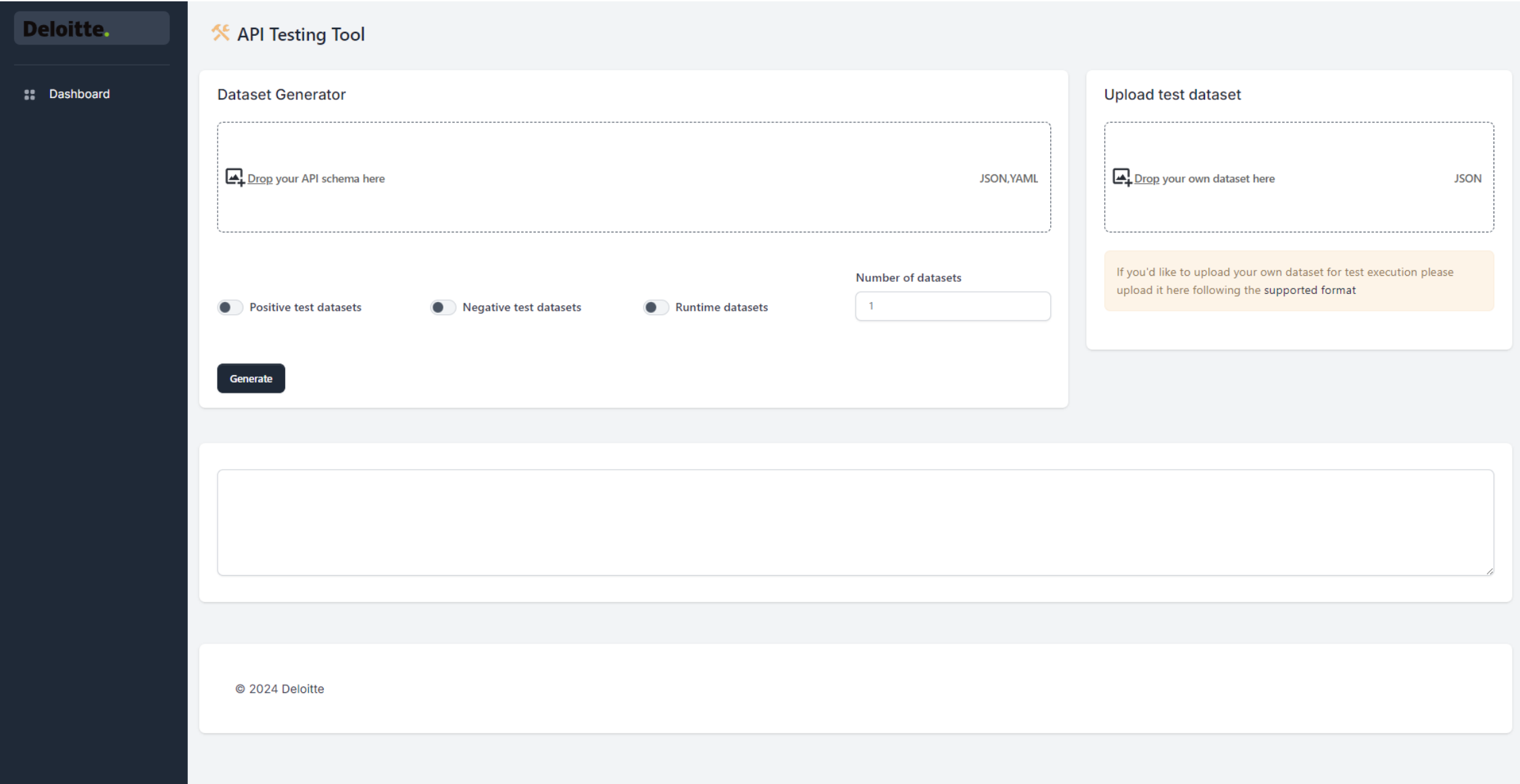
Recommended installation:

- Docker CLI / Desktop

Manual installation:

- Python (3.9 or above)
- Node (20 alpine)
- Yarn

Gen AI API Testing Tool - Screenshots



Gen AI API Testing Tool – Screenshots – Generated Test data

Deloitte.

Dashboard

API Testing Tool

Dataset Generator

Uploaded Successfully! Upload another?

JSON,YAML

Positive test datasets

Negative test datasets

Runtime datasets

Number of datasets

2

Generate

Upload test dataset

Drop your own dataset here

JSON

If you'd like to upload your own dataset for test execution please upload it here following the supported format

Testcase Executor

Optional Parameters

Token

Your token here

Test Dataset

```
{
  "http://localhost:9097/api/getCurrency": {
    "method": "Get",
    "negative": [
      {
        "description": "Fetching currency data when service is down",
        "expectedResponse": {
          "code": "404",
          "message": "Service Not Found"
        }
      }
    ]
  }
}
```

Execute

Export as JSON

Gen AI API Testing Tool – Screenshots – Test case execution

Deloitte.

Dashboard

Testcase Executor

Export as JSON

Optional Parameters

Token Bearer eyJhbGciOiJIUzI1NiIsImtpZCI6Imd6RWE5UmdEWGYwM3htcHpTZHBLOE1TcnMzOUZEcmlVCV54bU5uZThxTG8iLCJ0eXAiOiJKV1QiLCJ0eXciOiJlel

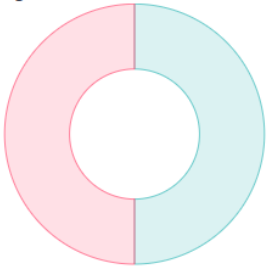
Test Dataset

```
{  "riskSelectedLob": "Fire",  "sortBy": "state",  "sortDirection": "desc"}],  "runtime": {}}
```

Execute

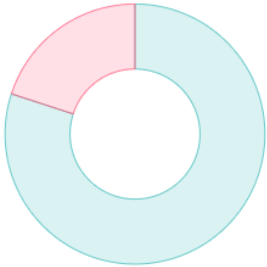
Result Summary

Negative Cases



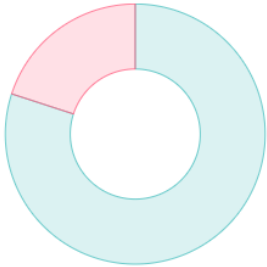
Success Failure

Positive Cases



Success Failure

Runtime Cases



Success Failure

Endpoint Test Report

Export as JSONExport as XLSX

ENDPOINT	CASE TYPE	STATUS	INFO
http://localhost:9097/api/getCurrency	negative	Failure	View Details

Gen AI API Testing Tool – Screenshots – Positive case success test report

Deloitte.

Dashboard

Success

Failure

Success

Failure

Success

Failure

Endpoint Test Report

Export as JSON

Export as XLSX

Endpoint	Case Type	Status	Info
http://localhost:9097/api/getCurrency	negative	Failure	View Details
http://localhost:9097/api/getCurrency	positive	Failure	View Details
http://localhost:9097/api/getCurrency	runtime	Failure	View Details
http://localhost:9097/api/getRiskDocumentsForOrderNumber	negative	Failure	View Details
http://localhost:9097/api/getRiskDocumentsForOrderNumber		Success	View Details
http://localhost:9097/api/getRiskDocumentsForOrderNumber		Success	View Details
http://localhost:9097/api/getRiskDocumentsForProgramLevel		Success	View Details
http://localhost:9097/api/getRiskDocumentsForProgramLevel		Success	View Details
http://localhost:9097/api/getRiskDocumentsForProgramLevel		Success	View Details
http://localhost:9097/api/locationsOrObjects/getNatCatWorldMapData		Success	View Details
http://localhost:9097/api/locationsOrObjects/getNatCatWorldMapData		Success	View Details
http://localhost:9097/api/locationsOrObjects/getNatCatWorldMapData	runtime	Success	View Details
http://localhost:9097/api/locationsOrObjectsWidgets/get	negative	Failure	View Details
http://localhost:9097/api/locationsOrObjectsWidgets/get	positive	Success	View Details
http://localhost:9097/api/locationsOrObjectsWidgets/get	runtime	Success	View Details

Detailed Report

Endpoint: http://localhost:9097/api/getRiskDocumentsForOrderNumber

Status: Success

Case Type: positive

Response:

```
{"cityFilterList":null,"countryFilterList":null,"documentTypeFilterList":null,"documentsResponseList":[],"locationNameFilterList":null,"resultCount":null,"totalCount":null}
```

Close

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Next steps

- *CI/CD integration of API testing tool via Jenkins*
- *Enhancing the tool to validate response for structural correctness*



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